

Department of Mathematics

Question Bank

Subject Name: Applied Discrete Mathematical Structures and Graph Theory

Subject Code: 22MAT41A/22MAT41S

Branch: AI & DS, AI & ML, CSE, ISE

Module - I

1. Define the propositions "Conjunction, Disjunction, Exclusive disjunction, Conditional biconditional, Negation" along with the truth tables.
2. Define Converse, Contrapositive and Inverse of the conditional proposition.
3. Construct the Truth tables for the following compound propositions.
 - (a) $p \wedge \neg p$
 - (b) $p \vee \neg p$
 - (c) $(p \vee \neg q) \rightarrow (p \wedge q)$
 - (d) $(p \rightarrow q) \vee (\neg p \rightarrow q)$
 - (e) $(p \rightarrow q) \wedge (\neg p \rightarrow r)$
4. Define Tautology, Contradiction, Contingency and Logical Equivalences.
5. Express each of the following propositions as an English sentence:
 - (a) Given p : I bought a lottery ticket this week, q : I won the million dollar jackpot.
 $\neg p, p \vee q, p \rightarrow q, p \leftrightarrow q, p \wedge q, \neg(p \wedge q)$.
 - (b) Given p : The election is decided, q : The votes have been counted.
 $\neg p \wedge q, \neg p \rightarrow \neg q, \neg q \vee (\neg p \wedge q)$.
6. Given p : You drive over 65 miles per hour. q : You get a speeding ticket. Translate the following into propositional logic:
 - (a) You do not drive over 65 miles per hour.
 - (b) You drive over 65 miles per hour, but you do not get a speeding ticket.
 - (c) You will get a speeding ticket if you drive over 65 miles per hour.
 - (d) If you do not drive over 65 miles per hour, then you will not get a speeding ticket.
7. Write the truth values of the following propositions:
 - (a) If $3 + 4 = 12$, then $3 + 2 = 6$.
 - (b) If Obama is the president of the United States, then $2 + 3 = 5$.
 - (c) $2 + 2 = 4$ or Monkey will fly in the sky.
 - (d) $2 + 2 = 4$ and Monkey will fly in the sky.
 - (e) $0 > 1$ if and only if $2 > 1$.
8. Write the converse, contrapositive and inverse of the following statements:
 - (a) If it is raining then the home team wins.

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- (b) If Bolt is concerned about his cholesterol levels, then he will walk at least two miles three times a week.
- (c) If I come to class, then there will be a quiz.
9. Show that the following expressions are logically equivalent by using truth table.
- (a) $\neg(p \vee q)$ and $\neg p \wedge \neg q$.
- (b) $p \rightarrow q$ and $\neg p \vee q$.
- (c) $\neg p \rightarrow (q \rightarrow r)$ and $q \rightarrow (p \vee r)$
10. Verify whether the following logical expressions are a tautology or not by using truth table.
- (a) $(p \wedge q) \rightarrow (p \rightarrow q)$
- (b) $[\neg p \wedge (p \vee q)] \rightarrow q$.
- (c) $[\neg p \wedge (p \vee q)] \rightarrow q$.
- (d) $q \leftrightarrow (\neg p \vee \neg q)$
11. Show that the following compound propositions are logically equivalent by developing a series of logical equivalences(Rules of logic).
- (a) $\neg[(p \vee q) \rightarrow r]$ and $(p \vee q) \wedge \neg r$
- (b) $(p \vee q) \wedge \neg(\neg p \wedge q)$ and p
- (c) $[p \vee q \vee (\neg p \wedge \neg q \wedge r)]$ and $(p \vee q \vee r)$
- (d) $(\neg p \vee \neg q) \rightarrow (p \wedge q \wedge r)$ and $p \wedge q$
- (e) $(p \rightarrow q) \wedge [\neg q \wedge (r \vee \neg q)]$ and $\neg(q \vee p)$
12. Establish the validity of the arguments by using rules of inference.
- (a) If Rahul hits a century, then he will get a car as gift.
Rahul will get a car as gift.
Therefore, Rahul hits a century.
- (b) I will become famous or I will not become a musician.
I will become a musician.
Therefore I will become famous.
- (c) If you have a current password, then you can log onto the network.
You have a current password.
Therefore, You can log onto the network.
- (d) If Roger studies, then he will pass examination.
If Roger doesn't play football, then he'll study.
Roger failed in examination.
Therefore Roger plays football.

- (e) If Ravi goes out with friends, then he will not study.
If Ravi does not study, then his father becomes angry.
His father is not angry.
Therefore Ravi has not gone out with friends.
- (f) If you send me an e-mail message, then I will finish writing the program.
If you do not send me an e-mail message, then I will go to sleep early.
If I go to sleep early, then I will wake up feeling refreshed.
Therefore If I do not finish writing the program, then I will wake up feeling re-freshed.
- (g) It is not sunny this afternoon and it is colder than yesterday.
If we go swimming, then it is sunny.
If we do not go swimming, then we will take a canoe trip.
If we take a canoe trip, then we will be home by sunset.
Therefore We will be home by sunset.
13. Express the following system specifications into logical expressions.
- (a) "The diagnostic message is stored in the buffer or it is re- transmitted."
"The diagnostic message is not stored in the buffer."
"If the diagnostic message is stored in the buffer, then it is re-transmitted."
- (b) "If the router can send packets to the edge system, then it supports the new address space."
"If the router supports the new address space, then the latest software release is installed".
"If the latest software release is installed, then the router can send packets to the edge system".
"The router does not support the new address space."
14. Express the following sentences into logical expressions by using predicates and quantifiers.
- (a) All lions are fierce.
(b) Some lions do not drink coffee.
(c) Some fierce creatures do not drink coffee.
15. Translate each of these statements into logical expressions using predicates, quantifiers, and logical connectives. Choose the relevant domain.
- (a) No one is perfect.
(b) Not everyone is perfect.
(c) All your friends are perfect.
(d) At least one of your friends is perfect.
(e) Everyone is your friend and is perfect.
16. Express the following sentences into logical expressions by using predicates and quantifiers.
- (a) Every mail message larger than one megabyte will be compressed.

- (b) If a user is active, at least one network link will be available.
- (c) Every student in your class has taken a course in calculus.
- (d) Every student in this class has visited either Canada or Mexico.

17. Problems on proof.

- (a) If n is an odd integer, then n^2 is odd (Direct Proof).
- (b) Prove that if $m + n$ and $n + p$ are even integers, where m , n , and p are integers, then $m + p$ is even (Direct Proof).
- (c) Show that the sum of two odd integers is even (Direct Proof).
- (d) For every integers n , if $3n + 2$ is an odd integer, then n is odd (Proof by contrapositive).
- (e) Prove that if m and n are integers and mn is even, then m is even or n is even (Proof by contrapositive).
- (f) Show that if n is an integer and $n^3 + 5$ is odd, then n is even (Proof by contradiction).
- (g) Prove that if n is an integer and n^2 is even, then n is even (Proof by contradiction).

Department of Mathematics

Question Bank

Module - II

1. Define: Relation, Reflexive, Symmetric, Antisymmetric, Transitive, Equivalence relation, Irreflexive, Zero-one matrix (Matrix representation), Digraph, In-degree and out-degree of vertices, Partial order Set (Poset), Lattices, Semi group, Monoid, Group, Abelian group, Subgroup, Homomorphism of groups.
2. Let $A = \{1, 2, 3, 4\}$ and let $R = \{(1, 1), (1, 4), (4, 2), (2, 1), (2, 4), (3, 1), (3, 4), (4, 2)\}$ be a relation defined on A .

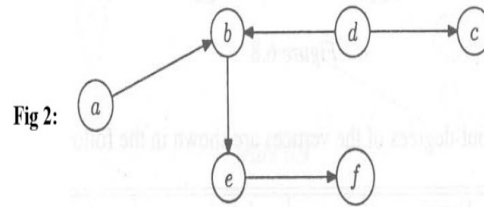
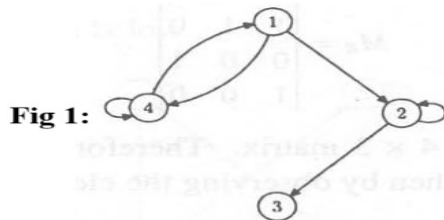
- (a) Draw the digraph of R and hence determine the in-degree and out-degree of the vertices.
- (b) Represent R as a zero-one matrix.

3. Determine the relations from the following matrix representations:

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, M_S = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

If there exist digraphs, represent the same and determine the in-degree and out-degree of the vertices.

4. Find the relation represented by the digraph given below, and write its matrix representation. Also, determine the in-degree and out-degree of the vertices.



5. Let $A = \{1, 2, 3, 4\}$. Let $R = \{(1, 1), (1, 2), (2, 3), (3, 3), (3, 4)\}$, and $S = \{(1, 1), (1, 3), (1, 4), (2, 2), (3, 2), (4, 4)\}$ be the relations on A . Find the relations $R \circ S$, $S \circ R$, R^2 , S^2 and their respective matrices. Draw the digraphs also. Also, draw the respective digraphs and represent the matrices.

6. Let $A = \{a, b, c\}$, and let R, S be relations on A whose matrices are given below:

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}, M_S = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Find the relations $R \circ S$, $S \circ R$, R^2 , S^2 and their respective matrices. Draw the digraphs also.

7. Determine whether the following relations are reflexive, irreflexive, symmetric, antisymmetric, transitive, and asymmetric. Also, verify which of the relations are equivalence relations.

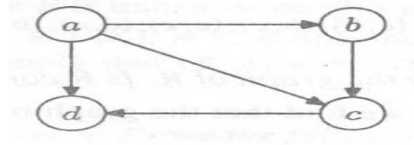
(a) $A = \{1, 2, 3, 4\}$, $R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$

(b) $R_2 = \{(1, 1), (1, 2), (2, 1)\}$

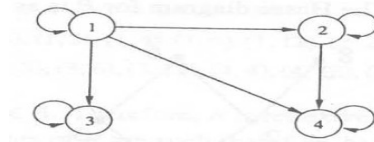
(c) $R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$

(d) $R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$

(e) $M_R = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ and $M_S = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$

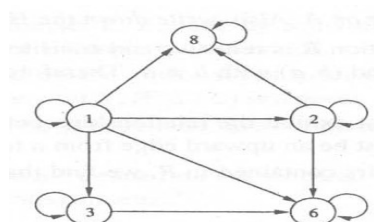


8. Let $A = \{1, 2, 3, 4, 5, 6\}$ be a set. Define a relation R by $(x, y) \in R$ if and only if $x - y$ is a multiple of 3. Verify that R is an equivalence relation.
9. Suppose that R is the relation on the set of strings of English letters such that aRb if and only if $l(a) = l(b)$, where $l(x)$ is the length of the string x . Is R an equivalence relation?
10. Let R be the relation on the set of all URLs (or Web addresses) such that xRy if and only if the Web page at x is the same as the Web page at y . Show that R is an equivalence relation.
11. Show that the relation R consisting of all pairs (x, y) such that x and y are bit strings of length three or more that agree in their first three bits is an equivalence relation on the set of all bit strings of length three or more.
12. Let R be a relation on the set $A = \{1, 2, 3, 4\}$ defined by xRy if and only if x divides y . Prove that (A, R) is a poset and draw its Hasse diagram.
13. Let $A = \{1, 2, 3, 4, 5\}$ be a set. Define a relation R by xRy if and only if $x \geq y$. Prove that R is a poset and draw its Hasse diagram.
14. Let $A = \{1, 2, 3, 4\}$ with the following digraph representing the relation R :



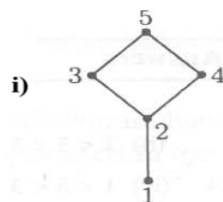
Verify R is a poset and draw its Hasse diagram if it exists.

15. Let $A = \{1, 2, 3, 6, 8\}$ with the following digraph representing the relation R :

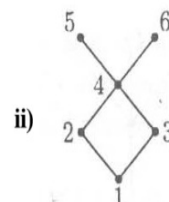


Verify R is a poset and draw its Hasse diagram if it exists.

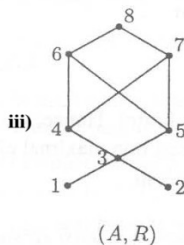
16. Is (S, R) a poset if S is the set of all people in the world and $(a, b) \in R$, where a and b are people, if a) a is taller than b ? b) a is not taller than b ? c) $a = b$ or a is an ancestor of b ? d) a and b have a common friend?
17. Draw the Hasse diagrams for the following posets and determine maximal, minimal and greatest, least elements.
 - (a) $(A, R) = (\{2, 4, 5, 10, 12, 20, 25\}, |)$
 - (b) All positive divisors of 36.
18. Let $A = 1, 2, 3, 4$ and $R = \{(1, 1), (1, 2), (2, 2), (2, 4), (1, 3), (3, 3), (3, 4), (1, 4), (4, 4)\}$. Verify R is a partial order on A and also draw the Hasse diagram for R .
19. Consider the Hasse diagrams of the posets (A, R) given below. Determine the, Least element, Greatest element, Maximal elements, Minimal elements. Also find All upper bounds, All lower bounds, Greatest lower bound (GLB) and Least upper bound (LUB) of the subset B of the set A . Also determine, whether these posets are lattices.



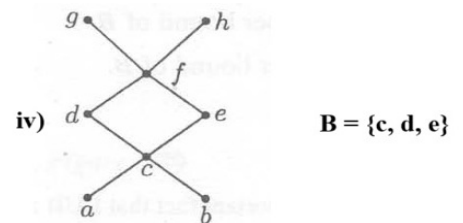
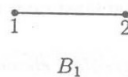
$B = \{2, 3, 4\}$



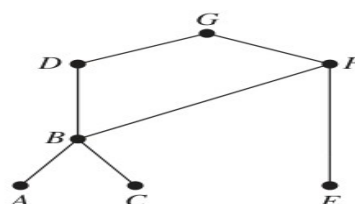
Where $B = \{1, 2, 3\}$



B_1 & B_2 are subsets

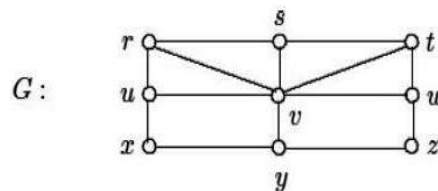


20. Find a compatible total ordering for the poset $(\{1, 2, 4, 5, 12, 20\}, |)$.
21. A development project at a computer company requires the completion of seven tasks. Some of these tasks can be started only after other tasks are finished. A partial ordering on tasks is set up by considering task $X \prec Y$ if task Y cannot be started until task X has been completed. The Hasse diagram for the seven tasks, with respect to this partial ordering, is shown in Figure below. Find an order in which these tasks can be carried out to complete the project.



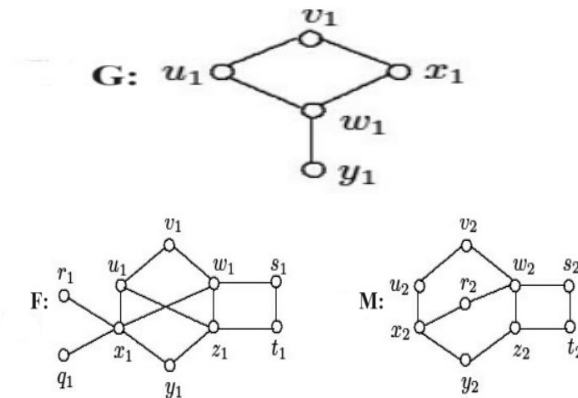
22. Show that the following are abelian groups:
- (a) Let G be a set of all non-zero real numbers and $a \times b = \frac{1}{2}ab$
 - (b) If $\circ$ is an operator on $\mathbb{Z}$ defined by $x \circ y = x + y + 1$
 - (c) Let G be the set of all real numbers not equal to -1 , $a * b = a + b + ab$
23. Prove that the set of all natural numbers $\mathbb{N}$ is a semi-group under usual addition. Also prove that $(\mathbb{N}, +)$ and $(\mathbb{N}, \times)$ are monoids.
24. Let $G = (\mathbb{Z}_6, +)$ and $H = \{0, 2, 4\}$. Prove that $(H, +)$ is a subgroup of G .
25. Let $f : (\mathbb{R}^+, \cdot) \rightarrow (\mathbb{R}, +)$ where $f(x) = \log_{10} x$. Prove that f is a homomorphism.
26. If $f : \mathbb{R} \rightarrow \mathbb{R}^+$ defined by $f(x) = e^x$ for all $x \in \mathbb{R}$, prove that f is a homomorphism.
27. State the necessary and sufficient conditions for the subgroup H of a group G .

1. Define Graph, walk, path, trail, connected graph, complete graph, sub graph, bipartite graph, isomorphic graphs, degree of a vertex.
2. Let $S = \{2, 3, 5, 8, 13, 21\}$. Draw the graph G whose vertex set is S and such that $ij \in E(G)$ for $i, j \in S$ if $i + j \in S$ or $|i - j| \in S$.
3. Let $S = \{2, 3, 4, 7, 11, 13\}$. Draw the graph G whose vertex set is S and such that $ij \in E(G)$ for $i, j \in S$ if $i + j \in S$ or $|i - j| \in S$.
4. The ten editors have decided on the seven committees: $c_1 = \{1, 2, 3\}$, $c_2 = \{1, 3, 4, 5\}$, $c_3 = \{2, 5, 6, 7\}$, $c_4 = \{4, 7, 8, 9\}$, $c_5 = \{2, 6, 7\}$, $c_6 = \{8, 9, 10\}$, $c_7 = \{1, 3, 9, 10\}$. They have set aside three time periods for the seven committees to meet on those Fridays when all ten editors are present. Some pairs of committees cannot meet during the same period because one or two of the editors are on both committees. Model this situation by graph.
5. Write the word graph for a collection of 3-letter English words, say ACT, ARC, ARM, ART, CAR, CAT, OAT, RAT. We say that a word W_1 can be transformed into a word W_2 if W_2 can be obtained from W_1 by performing exactly one of the following two steps: (1) interchanging two letters of W_1 ; (2) replacing a letter in W_1 by another letter. Model the graph such that one edge can be drawn if W_1 can be transformed to W_2 .
6. Let G be the graph as shown below, let $X = \{e, f\}$, where $e = ru$ and $f = vw$, and let $U = \{u, w\}$. Draw the sub graphs $G - X$ and $G - U$ of G .



7. For the above graph G give an example of each of the following or explain why no such example exists.
 - (a) An $x - y$ walk of length 6.
 - (b) A $v - w$ trail that is not a $v - w$ path.
 - (c) An $r - z$ path of length 2.
 - (d) An $x - z$ path of length 3.
 - (e) An $x - t$ path of length $d(x, t)$.
 - (f) A circuit of length 10.
 - (g) A cycle of length 8.
8. Draw the graphs $2K_4 \cup C_5 \cup P_3$ and $3P_4 \cup C_4 \cup K_4$

9. Determine whether the graphs G_1 and G_2 given below are bipartite. If a graph is bipartite, then redraw it indicating the partite sets; if not, then give an explanation as to why the graph is not bipartite.



10. For the following pairs G, H of graphs, draw $G + H$ and $G \times H$.
- $G = P_3$ and $H = K_2$.
 - $F = K_5$ and $S = K_2$.
11. If G is a graph of size m , then prove that $\sum_{v \in V(G)} \deg(v) = 2m$.
12. A certain graph G has order 14 and size 27. The degree of each vertex of G is 3, 4 or 5. There are six vertices of degree 4. How many vertices of G have degree 3 and how many have degree 5?
13. The degree of each vertex of a certain graph of order 12 and size 31 is either 4 or 6. How many vertices of degree 4 are there?
14. The degree of every vertex of a graph G of order 25 and size 62 is 3, 4, 5 or 6. There are two vertices of degree 4 and 11 vertices of degree 6. How many vertices of G have degree 5?
15. For the graph G given below, construct a 3-regular graph H containing G as an induced subgraph such that H has the smallest possible order. What is this order?



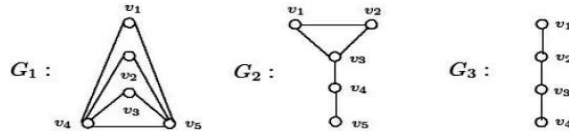
16. What is the minimum order of a 3-regular graph H containing the graph G in Figure below as an induced subgraph?



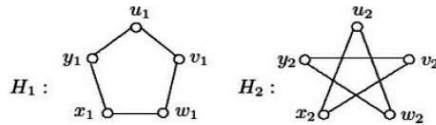
17. Determine which of the following sequences are graphical? For each of those that are graphical, construct a graph.

- $s_1 : 3, 3, 2, 2, 1, 1$
- $s_2 : 6, 5, 5, 4, 3, 3, 3, 2, 2$
- $s_3 : 7, 7, 4, 3, 3, 3, 2, 1$
- $s_4 : 6, 3, 3, 3, 3, 2, 2, 2, 1, 1$

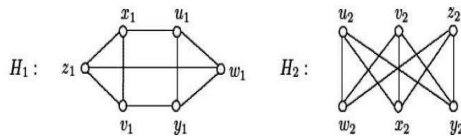
18. For the adjacency matrix A of the graph G_1, G_2 and G_3 of Figure below determine A^2, A^3 , without computing A or performing matrix multiplication.



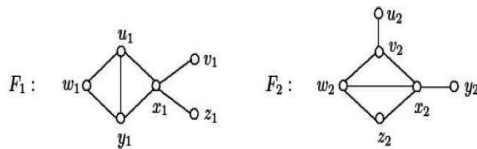
1. Determine whether the following graphs are Isomorphic.



(a)



(b)



(c)



(d)

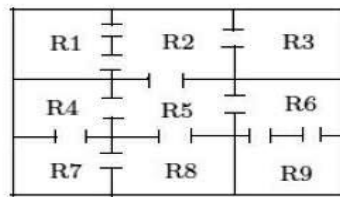
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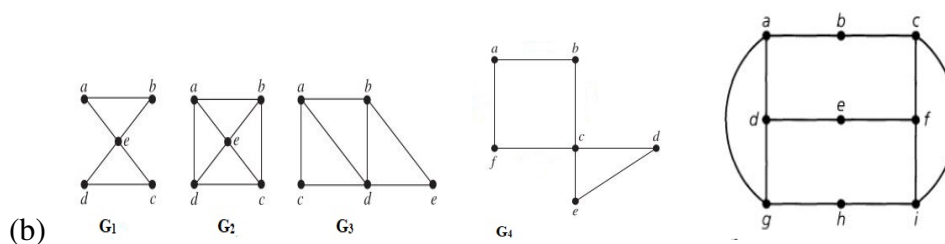
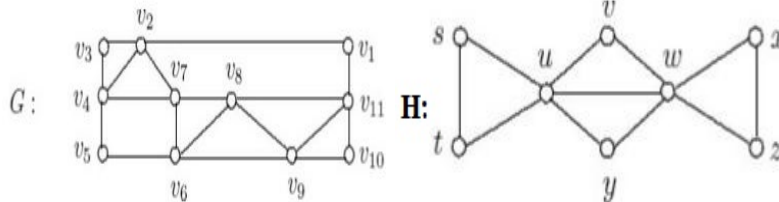
Applied Discrete Mathematical Structures and Graph Theory (22MAT41A)

Module - IV

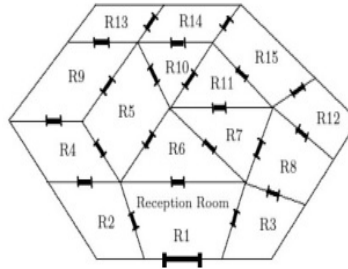
1. Define Eulerian Trail, Euler graph, Hamiltonian path and Hamiltonian cycle.
2. Discuss *Königsberg* bridge problem.
3. The diagram below shows the nine rooms on the second floor of a large house with doorways between various rooms. Is it possible to start in some room and go for a walk so that each doorway is passed through exactly once? How is this question related to graph theory?



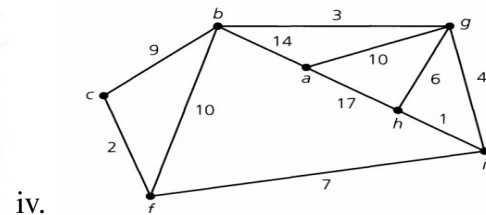
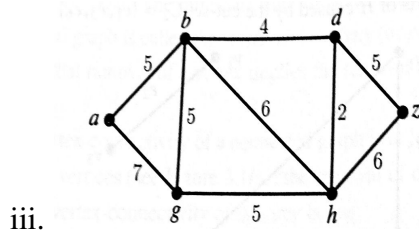
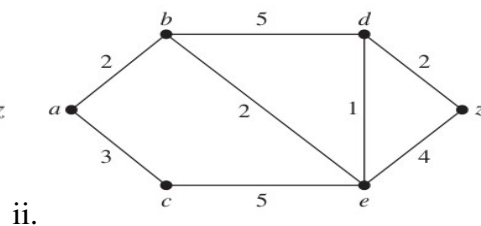
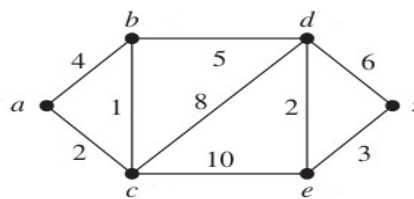
4. Determine whether the following graphs are Eulerian or Hamiltonian. If not, check whether the graph contains an Eulerian trail or Hamiltonian path.



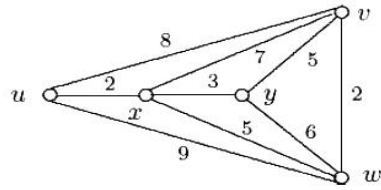
5. The following figure shows a diagram of a modern art museum that is divided into 15 exhibition rooms. At the end of each day, a security officer enters the reception room by the front door and checks each exhibition room to make certain that everything is in order. It would be most efficient if the officer could visit each room only once and return to the reception room. Can this be done?



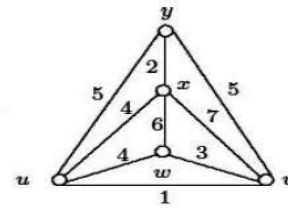
6. State traveling salesperson problem and its solution in Graph theory.
7. Use Dijkstra's algorithm to find the length of a shortest path between the vertices a and z in the following weighted graphs .



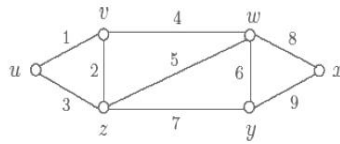
8. Define Tree and give the example for each of the tree of order 6, 7, 8, 10 which is not the path.
9. Prove that a graph G is a tree if and only if every two vertices of G are connected by a unique path.
10. Prove that every tree of order n has size $n - 1$.
11. The degrees of the vertices of a certain tree T of order 13 are 1, 2 and 5. If T has exactly three vertices of degree 2, how many end-vertices does it have?
12. A certain tree T of order 21 has only vertices of degree 1, 3, 5 and 6. If T has exactly 15 end - vertices and one vertex of degree 6, how many vertices of T have degree 5?
13. A certain tree T of order 35 is known to have 25 vertices of degree 1, two vertices of degree 2, three vertices of degree 4, one vertex of degree 5 and two vertices of degree 6. It also contains two vertices of the same (unknown) degree x . What is x ?
14. Explain Spanning tree and minimum spanning tree along with examples. Also state Kruskal's and Prim's Algorithms for finding minimum spanning tree.
15. Apply both Kruskal's and Prim's Algorithms, produces a minimum spanning tree in a connected weighted graphs given below.



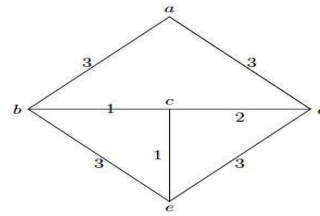
(a) i.



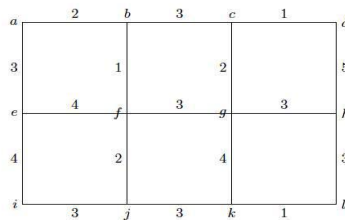
ii.



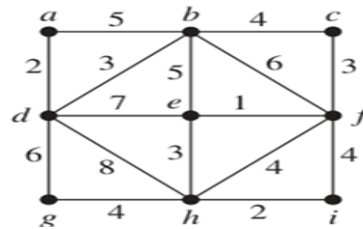
(b) i.



ii.

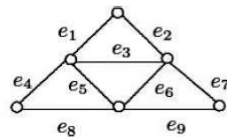


(c) i.

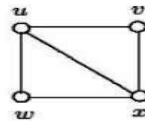


ii.

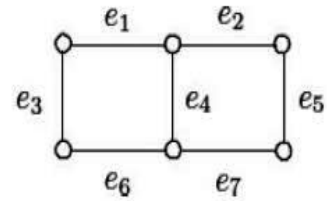
16. Determine the number of spanning trees of the following graphs.



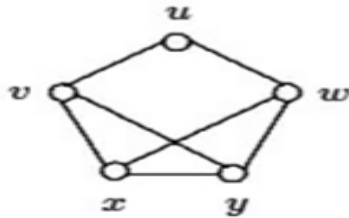
i.



ii.



iii.



iv.

Department of Mathematics

Question Bank

Applied Discrete Mathematical Structures and Graph Theory (22MAT41A)

Module - V

1. Define Matching, chromatic number and radio k – chromatic number, radio number ($rn(G)$) with examples.
2. As a result of doing well on an exam, six students Ashley (A), Bruce (B), Charles (C), Duane (D), Elke (E) and Faith (F) have earned the right to receive a complimentary text-book in either algebra (a), calculus (c), differential equations (d), geometry (g), history of mathematics (h), programming (p) or topology (t). There is only one book on each of these subjects. The preferences of the students are

A: d, h, t; B: g, p, t; C: a, g, h; D: h, p, t; E: a, c, d; F: c, d, p.

Can each of the students receive a book he or she likes?

3. Seven seniors Ben (B), Don (D), Felix (F), June (J), Kim (K), Lyle (L) and Maria (M) are looking for positions after they graduate. The University Placement Office has posted open positions for an accountant (a), consultant (c), editor (e), programmer (p), reporter (r), secretary (s) and teacher (t). Each of the seven students has applied for some of these positions:

B: c, e; D: a, c, p, s, t; F: c, r; J: c, e, r; K: a, e, p, s; L: e, r; M: p, r, s, t.

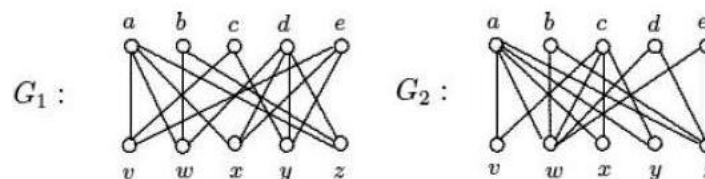
Is it possible for each student to be hired for a job for which he or she has applied?

4. There are positions open in seven different divisions of a major company: advertising (a), business (b), computing (c), design (d), experimentation (e), finance (f) and guest relations (g). Six people are applying for some of these positions, namely:

Alvin (A) : a, c, f; Beverly (B) : a, b, c, d, e, g; Connie (C) : c, f;
Donald (D) : b, c, d, e, f, g; Edward (E) : a, c, f; Frances (F) : a, f.

Represent this situation by a bipartite graph. Is it possible to hire all six applicants for six different positions? Give reason.

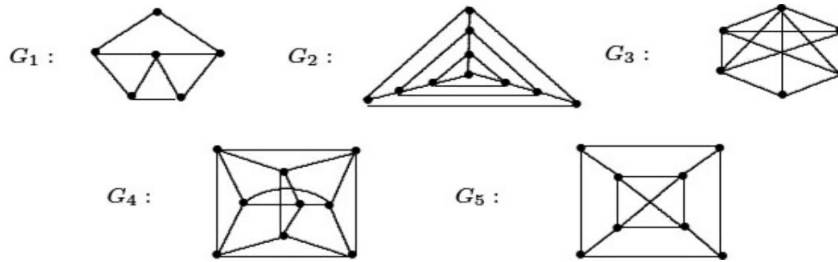
5. The following figure shows two bipartite graphs G_1 and G_2 , each with partite sets $U = \{v, w, x, y, z\}$ and $W = \{a, b, c, d, e\}$. In each case, can U be matched to W ?



6. Discuss the four color problem and state the four color theorem.

7. Discuss problem of five princes.

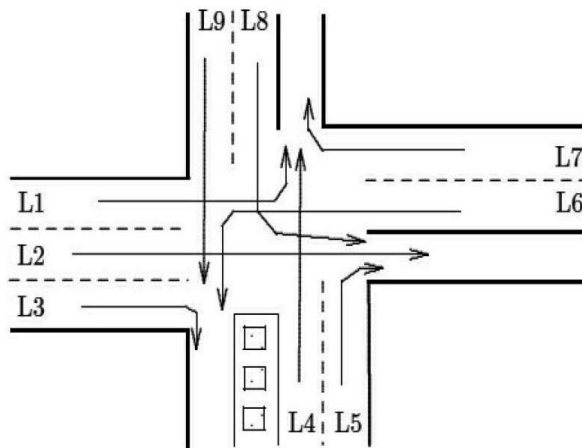
8. Find the chromatic number of the following graphs.



9. The mathematics department of a certain college plans to schedule the classes Graph Theory (GT), Statistics (S), Linear Algebra (LA), Advanced Calculus (AC), Geometry (G) and Modern Algebra (MA) this summer. Ten students (see below) have indicated the courses they plan to take. With this information, use graph theory to determine the minimum number of time periods needed to offer these courses so that every two classes having a student in common are taught at different time periods during the day. Of course, two classes having no students in common can be taught during the same period.

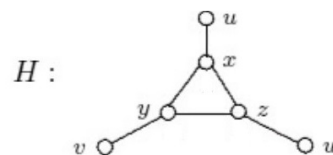
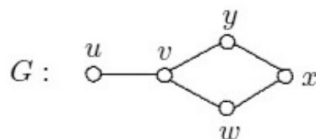
Anden: LA, S; Brynn: MA, LA, G; Chase: MA, G, LA; Denise: G, LA, AC; Everett: AC, LA, S; Fran ois: G, AC; Greg: GT, MA, LA; Harper: LA, GT, S; Irene: AC, S, LA; Jennie: GT, S.

10. The following figure shows the traffic lanes $L_1, L_2, \dots, L_9$ at the intersection of two busy streets. A traffic light is located at this intersection. During a certain phase of the traffic light, those cars in lanes for which the light is green may proceed safely through the intersection. What is the minimum number of phases needed for the traffic light so that (eventually) all cars may proceed through the intersection?



11. Ten editors have formed seven committees: $c_1 = \{1, 2, 3\}$, $c_2 = \{1, 3, 4, 5\}$, $c_3 = \{2, 5, 6, 7\}$, $c_4 = \{4, 7, 8, 9\}$, $c_5 = \{2, 6, 7\}$, $c_6 = \{8, 9, 10\}$, $c_7 = \{1, 3, 9, 10\}$. These committees are scheduled to meet every Friday, provided all ten editors are available. However, certain pairs of committees cannot meet simultaneously because they share one or more editors. Model this situation using graph theory and determine the minimum number of time schedules required to schedule all the meetings such that no editor has more than one meeting at the same time.

12. Eight chemicals are to be shipped across country by air express. The cost of doing this depends on the number of containers shipped. The cost of shipping one container is \$125. For each additional container the cost increases by \$85. Some chemicals interact with one another and it is too risky to ship them in the same container. The chemicals are labeled by $c_1, c_2, \dots, c_8$ and chemicals that interact with a given chemical are given below: What is the minimum cost of shipping the chemicals and how should the chemicals be packed into containers?
- $C_1 : C_2, C_5, C_6$ $C_2 : C_1, C_3, C_5, C_7$ $C_3 : C_2, C_4, C_7$
 $C_4 : C_3, C_6, C_7, C_8$ $C_5 : C_1, C_2, C_6, C_7, C_8$ $C_6 : C_1, C_4, C_5, C_8$
 $C_7 : C_2, C_3, C_4, C_5, C_8$ $C_8 : C_4, C_5, C_6, C_7$
13. Alvin (A) has invited three married couples to his summer house for a week: Bob (B) and Carrie (C) Hanson, David (D) and Edith (E) Irwin and Frank (F) and Gena (G) Jackson. Since all six guests enjoy playing tennis, he decides to set up some tennis matches. Each of his six guests will play a tennis match against every other guest except his/her spouse. In addition, Alvin will play a match against each of David, Edith, Frank and Gena. If no one is to play two matches on the same day, what is a schedule of matches over the smallest number of days?
14. Five individuals have been invited to a bridge tournament (bridge is a game of cards): Allen (A), Brian (B), Charles (C), Doug (D), Ed (E). A game of bridge is played between two 2-person teams. Every 2-person team $\{X, Y\}$ is to play against all other 2-person teams $\{W, Z\}$, where, of course, neither W nor Z is X or Y . If the same team cannot play bridge more than once on the same day, then what is the fewest number of days needed for all possible games of bridge to be played. Set up a schedule for doing this in the smallest number of days. What graph models this situation?
15. Seven softball teams from Atlanta, Boston, Chicago, Denver, Louisville, Miami and Nashville have been invited to participate in tournaments, where each team is scheduled to play a certain number of the other teams (given below). No team is to play more than one game each day. Set up a schedule of games over the smallest number of days.
- Atlanta : Boston, Chicago, Miami, Nashville
 Boston : Atlanta, Chicago, Nashville
 Chicago : Atlanta, Boston, Denver, Louisville
 Denver : Chicago, Louisville, Miami, Nashville
 Louisville : Chicago, Denver, Miami
 Miami : Atlanta, Denver, Louisville, Nashville
 Nashville : Atlanta, Boston, Denver, Miami
16. Determine $rn(G)$ for the graphs given below.



17. Determine $rn(K_3)$, $rn(K_4)$, $rn(C_5)$, $rn(C_6)$.